

Jiaheng Wang

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EMPLOYMENT

- Postdoctoral researcher University of Helsinki & HIIT, 2026 - ?
- Postdoctoral researcher University of Regensburg, 2024 - 2026
- Postdoctoral researcher University of Edinburgh, 2023 - 2024

EDUCATION

- Ph.D. University of Edinburgh, 2020 - 2023
Thesis: *Algorithms and complexity for approximately counting hypergraph colourings and related problems*
Advisor: Heng Guo
- B.Sc. *summa cum laude* in Computer Science (*Turing Class*) Peking University, 2016 - 2020

RESEARCH VISITING (FOR A LONG PERIOD)

- IT University of Copenhagen / BARC 07/2023 - 08/2023
Host: Radu Curticapean
- University of Oxford 06/2023
Host: Andreas Galanis and Leslie Ann Goldberg
- Queen Mary, University of London 06/2022
Host: Mark Jerrum
- Shanghai University of Finance and Economics 05/2020 - 09/2020
Host: Pinyan Lu
- University of Edinburgh 07/2019 - 08/2019
Host: Heng Guo
- Institute of Computing Technology, Chinese Academy of Sciences 09/2018 - 01/2020
Host: Xiaoming Sun

RESEARCH INTERESTS

- General theoretical computer science, especially algorithms and complexity of counting problems.
- Discrete mathematics, including extremal combinatorics and probabilistic combinatorics.

RESEARCH ARTICLES

[Link to Google Scholar] — [Link to DBLP]

- [13] **Beyond Bilinear Complexity: What Works and What Breaks with Many Modes?**
Cornelius Brand, Radu Curticapean, Petteri Kaski, Baitian Li, Ian Orzel, Tim Seppelt and J. Wang.
submitted
arXiv: 2602.11975; ECCC report TR26-025
- [12] **Sink-free orientations: a local sampler with applications.**
Konrad Anand, Graham Freifeld, Heng Guo, Chunyang Wang and J. Wang.
[C] 29th International Conference on Randomization and Computation (RANDOM 2025)
arXiv: 2502.05877

Last update: April 1, 2026. Author lists are sorted in the alphabetical order by convention. [J]: Journal, [C]: Conference.

- [11] **Can you link up with treewidth?**
Radu Curticapean, Simon Döring, Daniel Neuen and **J. Wang**.
[C] *42nd International Symposium on Theoretical Aspects of Computer Science (STACS 2025)*
arXiv: 2410.02606
- [10] **Rapid mixing of the flip chain over non-crossing spanning trees.**
Konrad Anand, Weiming Feng, Graham Freifeld, Heng Guo, Mark Jerrum and **J. Wang**.
[C] *41st International Symposium on Computational Geometry (SoCG 2025)*
arXiv: 2409.07892
- [9] **The complexity of computing fermionants and flow-like structures in graphs, modulo p .**
Isja Mannens and **J. Wang**.
submitted
- [8] **Approximate counting for spin systems in sub-quadratic time.**
Konrad Anand, Weiming Feng, Graham Freifeld, Heng Guo and **J. Wang**.
[J] *TheoretiCS, Volume 4 (2025), Article 3, 1–27*
[C] *51th International Colloquium on Automata, Languages and Programming (ICALP 2024)*
arXiv: 2306.14867
- [7] **Inapproximability of counting independent sets in linear hypergraphs.**
Guoliang Qiu and **J. Wang**.
[J] *Information Processing Letters, Volume 184, Article 106448, 1–6, 2024*
arXiv: 2212.03072
- [6] **Towards derandomising Markov chain Monte Carlo.**
Weiming Feng, Heng Guo, Chunyang Wang, **J. Wang** and Yitong Yin.
[J] *SIAM Journal on Computing, Vol. 54, Iss. 3, pp. 775–813, 2025*
[C] *64th IEEE Symposium on Foundations of Computer Science (FOCS 2023)*
arXiv: 2211.03487
- [5] **A simple polynomial-time approximation algorithm for the total variation distance between two product distributions.**
Weiming Feng, Heng Guo, Mark Jerrum and **J. Wang**.
[J] *TheoretiCS, Volume 2 (2023), Article 8, 1–7*
[C] *6th SIAM Symposium on Simplicity in Algorithms (SOSA 2023)*
arXiv: 2208.00740
- [4] **Swendsen-Wang dynamics for the ferromagnetic Ising model with external fields.**
Weiming Feng, Heng Guo and **J. Wang**.
[J] *Information and Computation, Volume 294, Article 105066, 1–34, 2023*
arXiv: 2205.01985
- [3] **Improved bounds for randomly colouring simple hypergraphs.**
Weiming Feng, Heng Guo and **J. Wang**.
[C] *26th International Conference on Randomization and Computation (RANDOM 2022)*
arXiv: 2202.05554
- [2] **Inapproximability of counting hypergraph colourings.**
Andreas Galanis, Heng Guo and **J. Wang**.
[J] *ACM Transactions on Computation Theory, 14(3–4):10, pp. 1–33, 2022*
arXiv: 2107.05486
- [1] **On the degree of Boolean functions as polynomials over $\mathbb{Z}_m$.**
Xiaoming Sun, Yuan Sun, **J. Wang**, Kewen Wu, Zhiyu Xia and Yufan Zheng.
[C] *47th International Colloquium on Automata, Languages and Programming (ICALP 2020)*
arXiv: 1910.12458

HONOURS AND AWARDS

- | | |
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| <ul style="list-style-type: none"> • Informatics Global PhD Scholarship (3.5 years) • 4 awards/scholarships during undergraduate study | <p>University of Edinburgh, 2020</p> <p>Peking University</p> |
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PROFESSIONAL / VOLUNTARY SERVICES

- Programme Committee member: FSTTCS'25.
- Layout editor of the journal TheoretiCS since 04/2025. (Voluntary. This is a Diamond Open Access journal published by the TheoretiCS Foundation e.V., a German not-for-profit organization. Please consider supporting it!)
- External reviewer at conferences: ICALP'26, STOC'26, STACS'26, SODA'26, MFCS'25, ESA'25, CIAC'25, APPROX/RANDOM'24, FOCS'24, ICALP'21, SODA'21.
- Journal referee: *Theoretical Computer Science*.
- Student organizer of SAGT'18 (organizing volunteers, getting involved in press, etc.).

TALKS

- Can you link up with treewidth?
 - IJTCS-FAW 2025 (invited Youth Forum talk), Paris, France, 07/2025
 - Highlight of Algorithms 2025, ETH Zürich, Switzerland, 06/2025
 - STACS 2025, Jena, Germany, 03/2025
 - LFCS Seminar, University of Edinburgh, United Kingdom. 11/2024
- Approximate counting for spin systems in sub-quadratic time
 - Peking University, Beijing, China. 09/2023
 - Shanghai Jiao Tong University, Shanghai, China. 09/2023
 - NII Shonan Meeting No. 186 “MCMC 2.0”, Kanagawa, Japan. 09/2023
- Towards derandomising Markov chain Monte Carlo
 - Basic Algorithm Research Copenhagen (BARC), Denmark. 08/2023
- A simple polynomial-time approximation algorithm for the total variation distance between two product distributions
 - University of Science and Technology of China, Hefei, China. 09/2023
 - QuACT classical talk, Beijing, China. 09/2023
 - Algorithms and Complexity Theory Seminars, Oxford, United Kingdom. 06/2023
 - LFCS Lab Lunch, Edinburgh, United Kingdom. 04/2023
 - SOSA 2023, Florence, Italy. 01/2023
- Improved bounds for randomly colouring simple hypergraphs
 - APPROX/RANDOM 2022, Champaign, IL, United States (virtual conference). 08/2022
 - Highlights of Algorithms 2022, LSE & QMUL, London, United Kingdom. 06/2022
- Inapproximability of counting hypergraph colourings
 - CS Peer Talk, Peking University, Beijing, China (virtual). 07/2022
 - Highlights of Algorithms, LSE & QMUL, London, United Kingdom. 06/2022
- On the degree of Boolean functions as polynomials over $\mathbb{Z}_m$.
 - ICALP 2020, Saarbrücken, Germany (virtual conference). 06/2020

TEACHING

- At University of Regensburg:
 - 70101 Complexity Theory

Durchführender, 2024 Winter

- At University of Edinburgh:
 - INFR08026 Introduction to Algorithms and Data Structures Teaching Assistant/Tutor, 2022/23
 - INFR11201 Randomized Algorithms Tutor, 2022 Autumn
 - INFR08026 Introduction to Algorithms and Data Structures Teaching Assistant/Tutor, 2021/22
- At Peking University:
 - 04834010 Randomized Algorithms Teaching Assistant, 2020 Spring
 - 04833440 Introduction to the Theory of Computation Teaching Assistant, 2020 Spring
 - 04833040/04832363 Introduction to Computer Systems Teaching Assistant/Tutor, 2019 Fall
 - 04833440 Introduction to the Theory of Computation Teaching Assistant, 2019 Spring
 - 04833040/04832363 Introduction to Computer Systems Teaching Assistant/Tutor, 2018 Fall